LAB6 REPORT: CODE AND BRANCH COVERAGE ANALYSIS

Aryaroop Bose - pes1ug24cs087

1. Objectives and Theoretical Principles

- This laboratory evaluates code coverage and branch coverage using pytest and coverage.py.
- Line (statement) coverage measures whether executable statements were executed at least once during test runs.
- However, line coverage can create a false sense of security by ignoring untested boolean branches within conditional structures (e.g. executing an if block without testing its corresponding False fallback).

Branch coverage requires every decision point (if, elif, compound boolean expressions) to evaluate to both True and False, ensuring that fallback paths, decision boundaries, and edge cases are systematically validated.

2. System Under Test: Smart Thermostat Controller (thermostat_control.py)

- The target module automates residential climate control through three core functions:
    - **target_temperature(mode, occupied, hour):** returns 16°C for 'away'; 18°C for nighttime 'sleep' (hour < 6 or >= 22) and 20°C for daytime 'sleep'; 18°C when unoccupied; 22°C during morning (6–9h); 21°C in evening (17–22h); and 20°C default daytime.
    - **hvac_action(current, target, outside):** computes diff = target - current. Returns 'idle' if abs(diff) < 1; 'heat_boost' if diff > 0 and outside < 5°C, else 'heat'; 'cool_boost' if diff < 0 and outside > 30°C, else 'cool'.

energy_alert(readings, limit): returns 'ok' if 0 readings exceed limit; 'critical' if >50% exceed limit; else 'warning'.

3. Test Methodology and Coverage Implementation

- Testing was conducted across two distinct phases to evaluate the operational differences between statement and decision coverage:
    - **Phase 1:** CodeCoverageTest.py was constructed with 14 test cases to achieve the >=90% statement coverage milestone. Lines 24, 27, and 38 were intentionally omitted to isolate line execution from complete branch closure, achieving 91% module coverage (32/35 statements) and 95% total suite coverage.

Phase 2: BranchCoverageTest.py expanded to 21 targeted unit test cases under coverage run --branch, adding boundary edge tests to achieve 100% statement coverage and 100% branch coverage (28/28 branches).

4. Terminal Test Execution and Coverage Evidence

The terminal execution captures for Phase 1 line coverage and Phase 2 branch coverage are shown side-by-side below:

```
(venv) $> coverage run -m pytest CodeCoverageTest_thermostat_control.py -v
========================================= test session starts ==========================================
platform linux -- Python 3.14.8, pytest-9.2.0, pluggy-1.7.0 -- /home/chthonic/repos/se-lab-pes1ug24cs087/l
ab6/L6_L7_29_thermostat_control/venv/bin/python3
cachedir: .pytest_cache
rootdir: /home/chthonic/repos/se-lab-pes1ug24cs087/lab6/L6_L7_29_thermostat_control
collected 14 items

CodeCoverageTest_thermostat_control.py::test_away_mode PASSED                                    [  7%]
CodeCoverageTest_thermostat_control.py::test_sleep_day PASSED                                    [ 14%]
CodeCoverageTest_thermostat_control.py::test_sleep_night_early PASSED                            [ 21%]
CodeCoverageTest_thermostat_control.py::test_sleep_night_late PASSED                             [ 28%]
CodeCoverageTest_thermostat_control.py::test_unoccupied PASSED                                   [ 35%]
CodeCoverageTest_thermostat_control.py::test_morning PASSED                                      [ 42%]
CodeCoverageTest_thermostat_control.py::test_evening PASSED                                      [ 50%]
CodeCoverageTest_thermostat_control.py::test_default_temperature PASSED                          [ 57%]
CodeCoverageTest_thermostat_control.py::test_idle PASSED                                         [ 64%]
CodeCoverageTest_thermostat_control.py::test_heat_boost PASSED                                   [ 71%]
CodeCoverageTest_thermostat_control.py::test_cool_boost PASSED                                   [ 78%]
CodeCoverageTest_thermostat_control.py::test_no_readings PASSED                                  [ 85%]
CodeCoverageTest_thermostat_control.py::test_all_within_limit PASSED                             [ 92%]
CodeCoverageTest_thermostat_control.py::test_warning PASSED                                      [100%]

========================================== 14 passed in 0.03s ==========================================
(venv) $> coverage report -m
Name                                    Stmts   Miss  Cover   Missing
---------------------------------------------------------------------
CodeCoverageTest_thermostat_control.py      30      0   100%
thermostat_control.py                       35      3    91%   24, 27, 38
---------------------------------------------------------------------
TOTAL                                       65      3    95%
(venv) $>
```

```
(venv) $> coverage erase
(venv) $> coverage run --branch -m pytest BranchCoverageTest_thermostat_control.py -v
========================================= test session starts ==========================================
platform linux -- Python 3.14.8, pytest-9.2.0, pluggy-1.7.0 -- /home/chthonic/repos/se-lab-pes1ug24cs087/l
ab6/L6_L7_29_thermostat_control/venv/bin/python3
cachedir: .pytest_cache
rootdir: /home/chthonic/repos/se-lab-pes1ug24cs087/lab6/L6_L7_29_thermostat_control
collected 21 items

BranchCoverageTest_thermostat_control.py::test_away_mode PASSED                                  [  4%]
BranchCoverageTest_thermostat_control.py::test_sleep_day PASSED                                  [  9%]
BranchCoverageTest_thermostat_control.py::test_unoccupied PASSED                                 [ 14%]
BranchCoverageTest_thermostat_control.py::test_morning PASSED                                    [ 19%]
BranchCoverageTest_thermostat_control.py::test_evening PASSED                                    [ 23%]
BranchCoverageTest_thermostat_control.py::test_default_temperature PASSED                        [ 28%]
BranchCoverageTest_thermostat_control.py::test_sleep_night_early PASSED                          [ 33%]
BranchCoverageTest_thermostat_control.py::test_sleep_night_late PASSED                           [ 38%]
BranchCoverageTest_thermostat_control.py::test_sleep_night_edge PASSED                           [ 42%]
BranchCoverageTest_thermostat_control.py::test_sleep_early_edge PASSED                           [ 47%]
BranchCoverageTest_thermostat_control.py::test_idle PASSED                                       [ 52%]
BranchCoverageTest_thermostat_control.py::test_heat PASSED                                       [ 57%]
BranchCoverageTest_thermostat_control.py::test_cool PASSED                                       [ 61%]
BranchCoverageTest_thermostat_control.py::test_heat_boost PASSED                                 [ 66%]
BranchCoverageTest_thermostat_control.py::test_cool_boost PASSED                                 [ 71%]
BranchCoverageTest_thermostat_control.py::test_heat_edge PASSED                                  [ 76%]
BranchCoverageTest_thermostat_control.py::test_cool_edge PASSED                                  [ 80%]
BranchCoverageTest_thermostat_control.py::test_no_readings PASSED                                [ 85%]
BranchCoverageTest_thermostat_control.py::test_all_within_limit PASSED                           [ 90%]
BranchCoverageTest_thermostat_control.py::test_warning PASSED                                    [ 95%]
BranchCoverageTest_thermostat_control.py::test_critical PASSED                                   [100%]

========================================== 21 passed in 0.04s ==========================================
(venv) $> coverage report -m
Name                                    Stmts   Miss Branch BrPart  Cover   Missing
-----------------------------------------------------------------------------------
BranchCoverageTest_thermostat_control.py      44      0      0      0   100%
thermostat_control.py                         35      0     28      0   100%
-----------------------------------------------------------------------------------
TOTAL                                         79      0     28      0   100%
(venv) $>
```

5. Codes

CodeCoverageTest_thermostat_control.py

```
import pytest
from thermostat_control import target_temperature, hvac_action,
energy_alert


# ==================== target_temperature tests ====================

def test_away_mode():
    assert target_temperature("away", True, 12) == 16


def test_sleep_day():
    assert target_temperature("sleep", True, 10) == 20


def test_sleep_night_early():
    assert target_temperature("sleep", True, 3) == 18


def test_sleep_night_late():
    assert target_temperature("sleep", True, 23) == 18


def test_unoccupied():
    assert target_temperature("home", False, 12) == 18


def test_morning():
    assert target_temperature("home", True, 7) == 22


def test_evening():
```

```
    assert target_temperature("home", True, 19) == 21


def test_default_temperature():
    assert target_temperature("home", True, 12) == 20


def test_idle():
    assert hvac_action(20, 20.5, 10) == "idle"


def test_heat_boost():
    assert hvac_action(15, 20, 0) == "heat_boost"


def test_cool_boost():
    assert hvac_action(25, 20, 35) == "cool_boost"


def test_no_readings():
    assert energy_alert([], 10) == "ok"


def test_all_within_limit():
    assert energy_alert([5, 8, 9], 10) == "ok"


def test_warning():
    assert energy_alert([5, 12, 6, 15], 10) == "warning"
```

BranchCoverageTest_thermostat_control.py

```
import pytest
from thermostat_control import target_temperature, hvac_action,
energy_alert
```

```python
# ==================== target_temperature tests ====================

def test_away_mode():
    assert target_temperature("away", True, 12) == 16


def test_sleep_day():
    assert target_temperature("sleep", True, 10) == 20


def test_unoccupied():
    assert target_temperature("home", False, 12) == 18


def test_morning():
    assert target_temperature("home", True, 7) == 22


def test_evening():
    assert target_temperature("home", True, 19) == 21


def test_default_temperature():
    assert target_temperature("home", True, 12) == 20


def test_sleep_night_early():
    assert target_temperature("sleep", True, 3) == 18


def test_sleep_night_late():
    assert target_temperature("sleep", True, 23) == 18


def test_sleep_night_edge():
    assert target_temperature("sleep", True, 23) == 18
```

```python
def test_sleep_early_edge():
    assert target_temperature("sleep", True, 6) == 20


# ===================== hvac_action tests =====================

def test_idle():
    assert hvac_action(20, 20.5, 10) == "idle"


def test_heat():
    assert hvac_action(15, 20, 10) == "heat"


def test_cool():
    assert hvac_action(25, 20, 25) == "cool"


def test_heat_boost():
    assert hvac_action(15, 20, 0) == "heat_boost"


def test_cool_boost():
    assert hvac_action(25, 20, 35) == "cool_boost"


def test_heat_edge():
    assert hvac_action(15, 20, 5) == "heat"


def test_cool_edge():
    assert hvac_action(25, 20, 30) == "cool"


# ===================== energy_alert tests =====================

def test_no_readings():
    assert energy_alert([], 10) == "ok"
```

```
def test_all_within_limit():
    assert energy_alert([5, 8, 9], 10) == "ok"


def test_warning():
    assert energy_alert([5, 12, 6, 15], 10) == "warning"


def test_critical():
    assert energy_alert([12, 15, 6, 11], 10) == "critical"
```